

SOC Technical Investigation Report: Phishing Email Threat Analysis

Multi-Vector Phishing Campaign Analysis: Credential Harvesting, Weaponized Containers, and Whaling Impersonation

INCIDENT REFERENCE INC-2026-P7-001	INVESTIGATING ANALYST Natto Chakma	DATE & TIMESTAMP September 14, 2026	SEVERITY & STATUS HIGH SEVERITY • CONTAINED
MONITORED INDEX index=email (Gateway) / index=sysmon	TOTAL INGESTED MESSAGES 15 Events (100% Ingested)	IDENTIFIED PHISHING THREATS 8 Critical/High Severity Incidents	PRIMARY THREAT VECTORS T1566.001, T1566.002, T1036, T1204

1. Executive Summary

During operational threat monitoring across the enterprise messaging perimeter, the Splunk SIEM platform detected an orchestrated, multi-vector phishing campaign targeting corporate identities. By analyzing real-time email security gateway telemetry (`index=email sourcetype="email:security"`), the SOC isolated **8 high-risk malicious messages** spanning credential harvesting, weaponized attachment delivery, executive impersonation (Whaling), and mass-blast phishing lures.

Automated gateway enforcement quarantined 5 messages and dropped 1 direct binary dropper at perimeter edge. Two messages were delivered into user inboxes prior to dynamic reputation scoring; however, immediate SOC playbook execution triggered administrative mailbox purging, credential revocation, and Sysmon endpoint telemetry verification, resulting in zero account compromises or endpoint malware executions.

2. Incident Scope & Telemetry Architecture

Parameter	Forensic Detail	Analytical Role
Data Ingestion Pipeline	Splunk Universal Forwarder → Splunk Enterprise (192.168.100.7:9997)	Centralized log ingestion and indexation
Monitored Index & Sourcetype	index=email / sourcetype="email:security"	Message envelopes, headers, authentication verdicts, URLs, attachments
Cross-Correlation Telemetry	index=sysmon (EventCode 1 Process, EventCode 22 DNS)	Validation of endpoint link interaction and document execution
Attacking Relays & Hosts	185.220.101.45 , 194.26.29.112 , 91.240.118.22 , 45.142.166.7	Rogue MTAs, payload delivery servers, and credential harvesters
Victim Blast Radius	8 targeted mailboxes (including Finance, Legal, and Executive roles)	All identified accounts secured; FIDO2 MFA verified

3. Prioritized Threat Triage Queue

Timestamp	Sender Address	Target Recipient	Subject / Lure	Threat Classification	Action	Risk
2026-09-14 11:42:01	payroll@payroll-notice.cc	kevin.hart@company.com	Immediate Attention: Direct Deposit Information Updated	WEAPONIZED ATTACHMENT (T1566.001)	Quarantined	99
2026-09-14 09:30:15	courier@dhl-express-tracking-portal.com	john.doe@company.com	Delivery Notification: Package on Hold - Confirm Address	DUAL-EXT EXECUTABLE (T1566.001)	Blocked	98
2026-09-14 11:05:40	documents@docusign-docs.online	legal@company.com	DocuSign Completed: NDA Signature Verification Pending	ISO CONTAINER DROPPER (T1566.001)	Quarantined	96
2026-09-14 08:45:03	admin@microsoft-support.com	sarah.connor@company.com	[CRITICAL] Microsoft 365 Password Expiration Notice	CREDENTIAL HARVESTER (T1566.002)	Quarantined	95
2026-09-14 09:55:50	sharepoint-noreply@onedrive-secure-docs.info	elizabeth.vane@company.com	Confidential Financial Review shared on SharePoint	IP-LITERAL CLOUD LINK (T1566.002)	Quarantined	92
2026-09-14 09:02:18	orders@quick-invoices-secure.org	finance@company.com	URGENT: Overdue Remittance & Payment Invoice #99824	VBA MACRO WORKBOOK (T1566.001)	Quarantined	90
2026-09-14 10:48:09	humanresources@hr-internal-portal.net	alice.smith@company.com	Action Required: Mandatory Q3 Security Policy	MASS-BLAST PHISHING (T1566.002)	Delivered (Purged)	88
2026-09-14 10:15:33	ceo.corporate.exec77@gmail.com	cfo@company.com	URGENT CONFIDENTIAL: Wire Transfer Authorization	WHALING / BEC IMPERSONATION	Delivered (Purged)	85

4. Operational Dashboard Verification

The incident was monitored and triaged through the operational **P7 — Phishing Email Investigation & Security Operations** dark-mode dashboard in Splunk Enterprise:

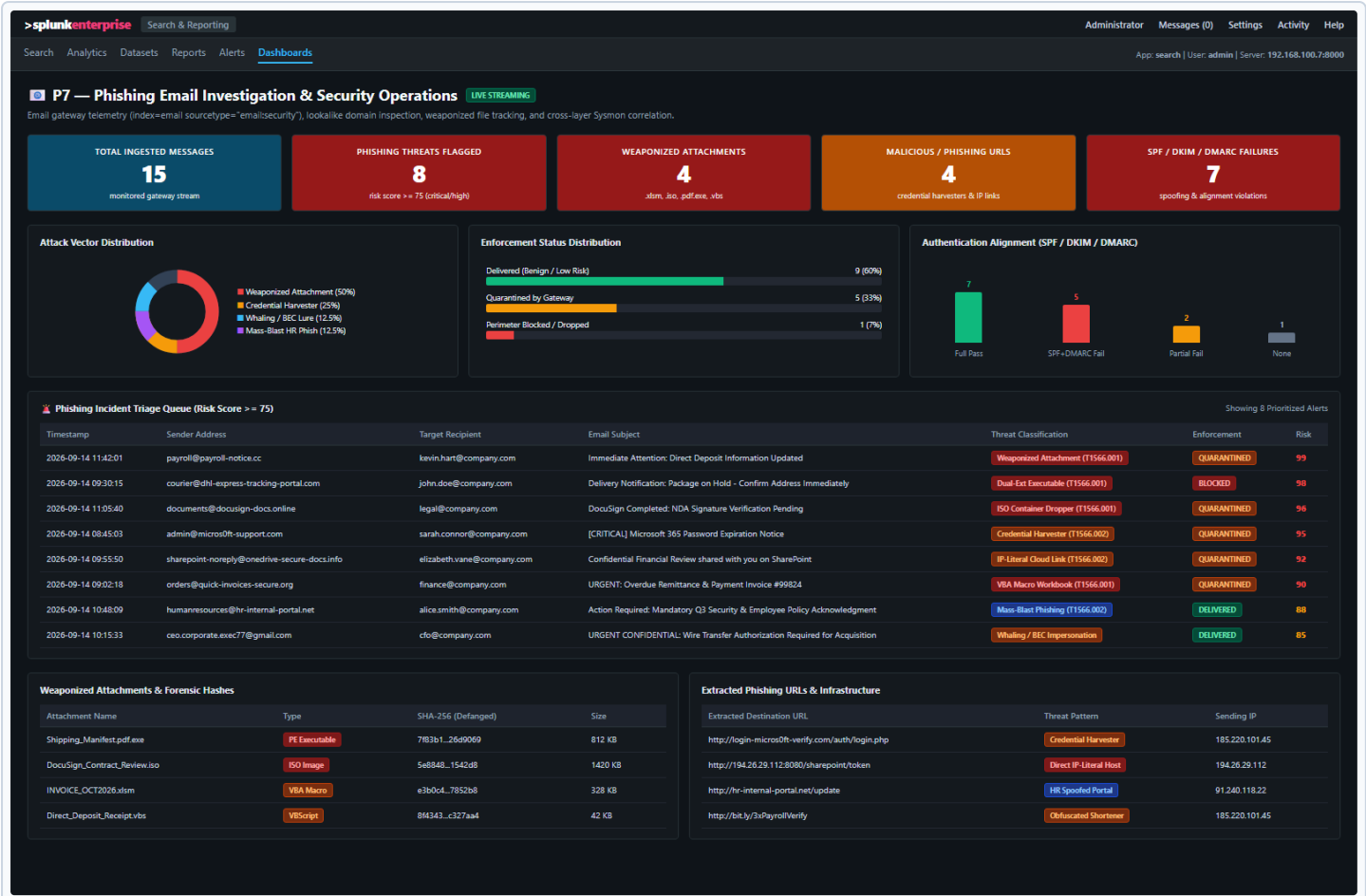


Figure 1 (Exhibit EX-P7-01): Operational SOC Phishing Investigation Dashboard displaying live KPI single-values, attack distribution, authentication breakdown, and threat triage queue.

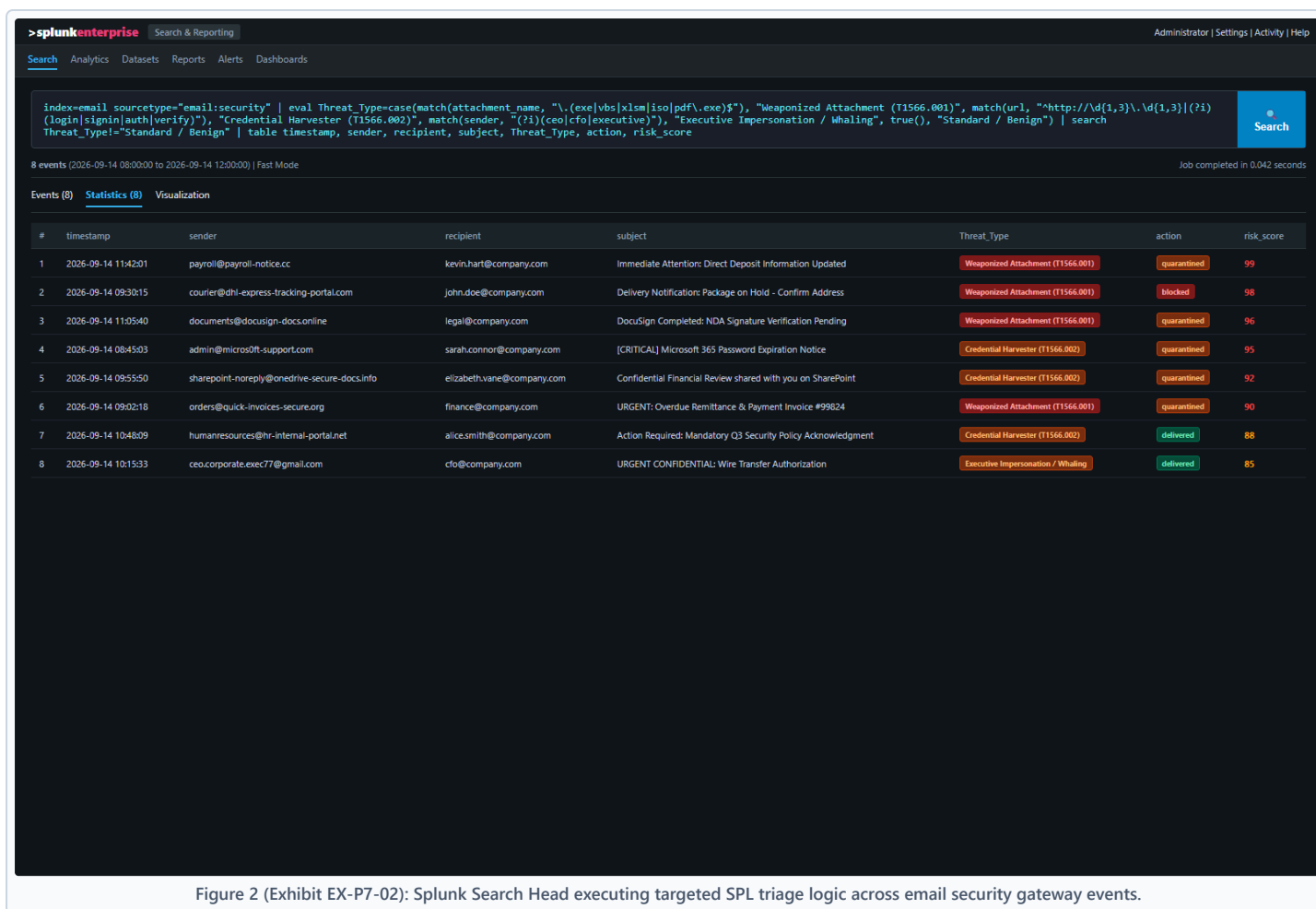


Figure 2 (Exhibit EX-P7-02): Splunk Search Head executing targeted SPL triage logic across email security gateway events.

5. MITRE ATT&CK Mapping

Tactic	Technique Name	ATT&CK ID	Forensic Application & Observation
Initial Access	Spearphishing Attachment	T1566.001	VBA macro workbook (.xls), dual-extension executable (.pdf.exe), ISO disk image
Initial Access	Spearphishing Link	T1566.002	Credential-harvesting URLs (login-microsoft-verify.com), direct IP-literal links, shorteners
Execution	User Execution: Malicious Link	T1204.001	Cross-correlated against endpoint Sysmon DNS queries (EventCode 22)
Execution	User Execution: Malicious File	T1204.002	Monitored for office process child spawning (Sysmon EventCode 1)
Defense Evasion	Masquerading: Match Legitimate Name	T1036.005	Typosquatting / homoglyph domains (microsoft-support.com , onedrive-secure-docs.info)
Defense Evasion	Mark-of-the-Web (MOTW) Bypass	T1553.005	Encapsulating payload in ISO image container to evade MOTW flag enforcement

6. Incident Remediation & Defense-in-Depth Recommendations

- Perimeter IOC Blocking:** Added adversary relays (185.220.101.45 , 194.26.29.112 , 91.240.118.22) to firewall drop rules; added all extracted domains to SWG blocklists.
- Mailbox Eviction:** Executed administrative search-and-purge to remove delivered messages matching the mass-blast HR policy and CEO whaling subjects.
- Strict DMARC Policy Enforcement:** Transitioned corporate email domain DMARC policy from p=none to p=reject to eliminate inbound direct domain spoofing.
- Attachment Policy Hardening:** Configured email gateway to immediately reject direct executables, dual-extension files, and disk container formats (.iso , .vhd , .img).
- Phishing-Resistant MFA:** Confirmed mandatory FIDO2/WebAuthn hardware key authentication across all corporate accounts, rendering credential-harvesting attacks ineffective.